Assignment 3 – Python programming (Object Oriented Programming)

1) Write a class Factorial for computing factorial of a number.

User of your class should be able to call:

fobj = Factorial(20) (here Factorial is a class name) and store the factorial in a data member called result

print(" result is: " fobj.result)

2) Write a class FileOperation which reads a file and has a method for each of the following tasks:

a) convert the contents of the file to upper case.
b) toggle the case of the contents of the file.
c) count of total number of words in the file
d) count of total number of alphanumeric characters in the file

3) Write a class MyString which inherits the string class (str) and implements a method for each of the following tasks:

a) check whether the given string is a palindrome
b) the number of words present in the string
c) count the non- alphanumeric characters

4) Fill up the implementations for the given class:

class BankAccount:

def __init__(self, initial_bal):

def deposit(self, amount):

def withdraw(self, amount):

def _generate_account_number(self): # Randomly generate a number between 1000 and 10,000 to be set only once, it cannot be changed thereafter. Add this logic while implementing this method

def transfer(self, amount, other_acc):

def get_account_number(self):

def set_owner(self, owner): # Name of owner to be set only once, it cannot be changed thereafter. Add this logic while implementing this method

def display(self): # display owner, acc_number and acc balance

def get_transaction_history(self): # get the history of the transactions done so far

```
acc = BankAccount(500)
acc.deposit(200)
acc.set_owner('Abhik')
acc.transfer(100,2560)
acc.withdraw(400)
acc.display()
acc.withdraw(400)    #should not work as balance is insufficient
acc.set_owner('Anand')  # should not change the name
acc.get_transaction_history()
```

5) Write a class to implement binary search functionality on lists.

Expected usage by the end user of your class:

```
Arr = [34,5,75,2,1,67]
bs = BinarySearch(arr)
bs.search(5)
```